

Inference

✓ Inference server

Gemma
VLLM

low
black
good

quantization

not quantized models

KvCache

Continuous batching

Speculative Decoding

✓ memory efficiency

latency ✓

throughput ✓

check

Safety margin



passed memory

Inference server

base model

Quantum

fundamental

- 1H-model

- 1P-model

CAS (12)

percentage



120

100% percentile

110

85% percentile

100

90

80

70

44% ~~percent~~

60

60

50

85
85
9

4
4
9

percentile(k), no. of values $\leq k$

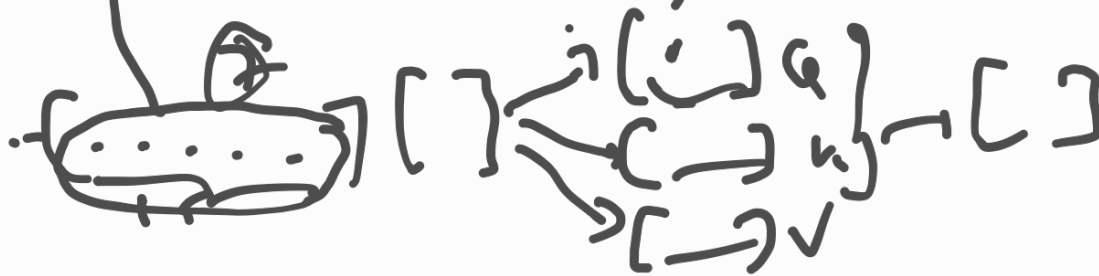
~~$\Theta(1) - \text{array}$~~

1

Kth Cache

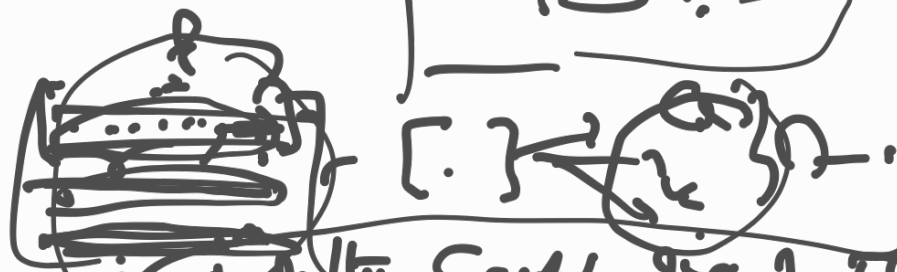
What is the Capital of India?

new



The

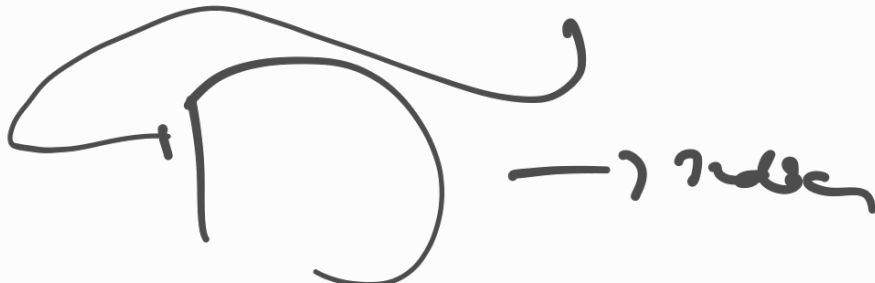
What is the center of mass? The



~~What is the center of mass? The center~~



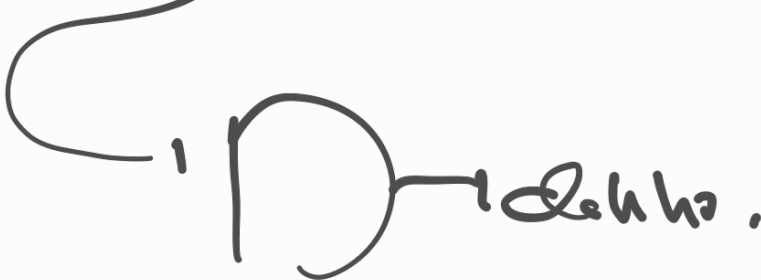
What is the center of mass? The center of



What is the center of mass? The center of mass



What is the center of mass? The center of mass



What is the center of mass? The center of mass

main efficiency

(11)

Agency movement

~~it's all over~~

Accounting

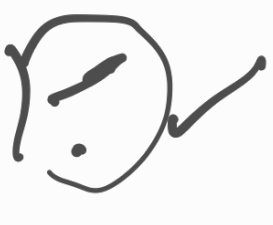
Power quality

Function

Interfacing

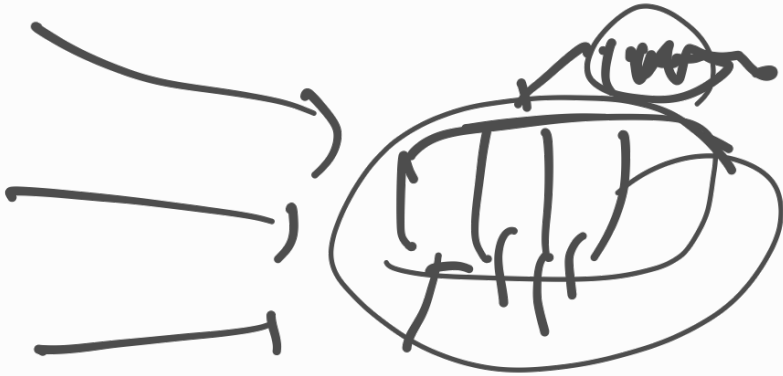
Control

K.V. Control

It can't flush out queries
to enter query. 

(11)

Cache batching



batch

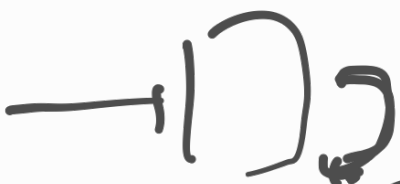
flush



(flush)

(15)

Speculative decoder



speculate max then
one taken & future

(5)

Parallel memory (distributed system)



Print Cache

F. J. S. S. S.

Cov. ..

m

1. H_2O

A hand-drawn sketch of a face with a wide, toothy grin and a single dot for an eye. The drawing is simple and expressive, with thick black lines.

A handwritten diagram illustrating a neural network architecture. It features a horizontal sequence of nodes connected by lines. The nodes are represented by circles and ovals. Some nodes contain mathematical expressions or symbols, such as ω_1 , ω_2 , ω_3 , and ω_4 . There are also nodes with more complex expressions like $\frac{\omega_1 \omega_2 \omega_3}{\omega_4}$ and $\frac{\omega_1 \omega_2 \omega_3 \omega_4}{\omega_5}$. The diagram shows a flow from left to right, with some nodes having multiple incoming or outgoing connections. The overall structure suggests a sequence of operations or a feed-forward network.

Handwritten notes for 'Lecture 2' showing a diagram of a network structure. The diagram consists of five nodes: 'su' (top left), 'paltc' (top center), 'ner' (top right), 'v' (bottom center), and 'm' (bottom right). The nodes are connected by lines: 'su' is connected to 'paltc' and 'ner'; 'paltc' is connected to 'ner' and 'v'; 'ner' is connected to 'v' and 'm'; 'v' is connected to 'm'.